Section A. Python for data and GenAI

12 questions, Q1 to Q12, about 30 minutes. Every question shows real code. Read it line by line; the answer is what the code does, never in what it was meant to do.

Q1 *Predict the output*

What does this code print?

```
orders = ["120", "80", "200"]
total = 0
for o in orders:
    total = total + o
print(total)
```

A. 400, since the strings are converted to numbers first
B. 12080200, since + joins the strings together
C. TypeError, since int and str cannot be added
D. 0, since the loop body never runs at all

Q2 Spot the bug, if there is one

Kavya expected q1 to stay as [4, 9, 7]. The code prints 4 9. Which statement is right?

```
q1 = [4, 9, 7]
q2 = q1
q2.append(3)
print(len(q1), sorted(q1)[-1])
```

A. There is a bug: q2 = q1 should have copied the list and Python failed to, so the change leaked into q1
B. The code is right: q2 = q1 makes both names refer to one list; q2 = q1.copy() would give a separate one
C. append() cannot affect q1 because it was called on q2, so the printed 4 must come from somewhere else in the program
D. sorted() changed q1 in place, and that in-place sort is what added the fourth element to the list

Q3 Trace the flow (pseudocode)

This is pseudocode, not Python. What does it print?

```
SET running_total TO 0
SET best_day TO none
FOR EACH (day, revenue) IN [(Mon, 40), (Tue, 25), (Wed, 55), (Thu, 55)]
    running_total <- running_total + revenue
    IF best_day IS none OR revenue > best_revenue THEN
        best_day <- day
        best_revenue <- revenue
    END IF
END FOR
PRINT best_day, running_total
```

A. Thu 175
B. Wed 120
C. Thu 55
D. Wed 175

Q4 Predict the output

An LLM has labelled six support tickets. What does the code print?

```
labels = ["refund", "delivery", "refund", "refund", "delivery", "other"]
count = {}
for lab in labels:
    count[lab] = count.get(lab, 0) + 1
print(count["refund"], len(count))
```

A. 3 3
B. 3 6
C. 6 3
D. KeyError: 'refund'

Q5 Fix the code

The last line raises TypeError: list indices must be integers or slices, not str. Which replacement for the last line returns the text 'Refund approved'?

```
resp = {"choices": [{"message": {"role": "assistant", "content": "Refund approved"}}],
        "usage": {"prompt_tokens": 412, "completion_tokens": 9}}
text = resp["choices"]["message"]["content"]
```

A. `text = resp["choices"]["message"][0]["content"]`
B. `text = resp["choices"][0]["message"]["content"]`
C. `text = resp.choices.message.content`
D. `text = resp["choices"]["message"]["content"][0]`

Q6 Predict the output

What does this code print?

```
sales = [1200, 0, 900, None, 300]
clean = [s for s in sales if s]
print(sum(clean) / len(clean), len(sales) - len(clean))
```

A. 600.0 1
B. 800 2
C. TypeError, because None cannot be summed
D. 800.0 2

Q7 Which line

Kavya's calculator gives 833.33 as the Plus average for the same rows, but the function returns 833. Which line causes the difference?

```
1  def average_by_tier(rows):
2      totals = {}
3      counts = {}
4      for tier, amount in rows:
5          totals[tier] = totals.get(tier, 0) + amount
6          counts[tier] = counts.get(tier, 0) + 1
7      result = {}
8      for tier in totals:
9          result[tier] = totals[tier] // counts[tier]
10     return result
```

(A) Line 9: the // operator discards the fraction

B. Line 5: totals should be written with += amount

C. Line 6: counts should start from 1, not 0

D. No line: the calculator is showing extra decimals

Q8 Predict the output

What does this code print?

```
def tag_order(order_id, tags=[]):
    tags.append(order_id)
    return tags

a = tag_order("K-101")
b = tag_order("K-102")
print(a, b)
```

✓A. ['K-101'] ['K-102']

B. ['K-101'] ['K-101', 'K-102']

(C) ['K-101', 'K-102'] ['K-101', 'K-102']

D. The output depends on the Python version

tags = [100]
a ✓

b tags = [100]
= [100, 200]

tags = [100, 200]

Q9 Fix the code

The comprehension crashes with ValueError on 'n/a'. Anand wants every bad value skipped and counted, whatever it looks like. Which rewrite does that?

```
def to_amount(text):
    return float(text.replace(",", ""))

rows = ["1,200", "950", "n/a", "300"]
amounts = [to_amount(r) for r in rows]
```

A.
```
amounts, rejected = [], []
for r in rows:
    try:
        amounts.append(to_amount(r))
    except ValueError:
        rejected.append(r)
```

B.
```
amounts = []
for r in rows:
    if r not in ("n/a", "NA", ""):
        amounts.append(to_amount(r))
```

C.
```
amounts = []
try:
    for r in rows:
        amounts.append(to_amount(r))
except ValueError:
    print("bad row found")
```

D.
```
amounts = [float(r.replace(",", "")) for r in rows if r.strip()]
```

Q10 Spot the bug

The code should print the customer with the highest spend, but it prints ('Zara', 700). Why?

```
customers = [("Asha", 4200), ("Ravi", 9100), ("Zara", 700)]
top = sorted(customers)[-1]
print(top)
```

A. sorted() returns ascending order, so [-1] picks the smallest value rather than the largest

B. Tuples cannot be compared with each other, so the data must be a list of lists first

C. 700 is compared as text, and the string '7' sorts after '4' and '9'

D. sorted() orders tuples by their first element, the name, unless a key is given

Q11 Predict the output

Kavya builds a classification prompt from a template. What happens on the second line?

```
template = ('Classify the ticket as one of {labels}. '
            'Reply as JSON: {"label": ...}. Ticket: {ticket}')
prompt = template.format(labels="refund, delivery, other", ticket="My parcel never came")
```

A. The prompt is built with both placeholders filled and the JSON kept as written

B. KeyError, since the braces around "label" read as a third placeholder

C. SyntaxError, since double quotes cannot sit inside a single-quoted string

D. The prompt is built but {ticket} is left unfilled because it comes last

Q12 Fix the code

grid should be two independent rows of three zeros. After grid[0][1] = 7 it prints [[0, 7, 0], [0, 7, 0]]. Which first line gives [[0, 7, 0], [0, 0, 0]]?

```
grid = [[0] * 3] * 2
grid[0][1] = 7
print(grid)
```

A. `grid = [[0, 0, 0]] * 2`
B. `grid = list([[0] * 3] * 2)`
C. `grid = [[0] * 3 for _ in range(2)]`
D. `grid = [[0] * 3] * 2 and then grid = grid.copy()`

Section B. SQL for data and GenAI

8 questions. Q13 to Q20, about 20 minutes. The small tables are printed once and reused where a question says so. NULL means the value is missing.

Q13 Predict the result

The orders table below is used for Q13 to Q16 and Q18. What does this query return?

order_id	customer_id	tier	order_date	amount	status
1	C1	Plus	2026-04-02	1200	paid
2	C2	Basic	2026-04-03	NULL	paid
3	C1	Plus	2026-04-05	800	cancelled
4	C3	Student	2026-04-06	300	paid
5	C2	Basic	2026-04-09	500	paid

```
SELECT COUNT(*), COUNT(amount), SUM(amount)
FROM orders
WHERE status <> 'cancelled';
```

A. 4, 4, 2000
B. 4, 3, 2000
C. 5, 4, 2800
D. 4, 3, NULL

Q14 Fix the query

On Postgres this query returns 0 for the orders table, where one order in five is cancelled. Which version returns 0.2?

```
SELECT SUM(CASE WHEN status = 'cancelled' THEN 1 ELSE 0 END) / COUNT(*) AS cancel_rate
FROM orders;
```

A. `SUM(CASE WHEN status = 'cancelled' THEN 1 ELSE 0 END) * 1.0 / COUNT(*)`
B. `ROUND(SUM(CASE WHEN status = 'cancelled' THEN 1 ELSE 0 END) / COUNT(*), 2)`
C. `COUNT(status = 'cancelled') / COUNT(*)`
D. `AVG(status = 'cancelled')`

Q15 Predict the result

Against the orders table, what does this query return?

```
SELECT tier, COUNT(*) AS n
FROM orders
WHERE status = 'paid'
GROUP BY tier
HAVING COUNT(*) > 1;
```

A. Plus 2 and Basic 2
B. Basic 2, Plus 1 and Student 1
C. No rows, because HAVING must refer to the alias n
D. Basic 2 only

Q16 Spot the bug

Kavya expects 4 (five orders minus the one at 800) but gets 3. Why?

```
SELECT COUNT(*) FROM orders WHERE amount <> 800;
```

A. <> is not valid in Postgres, so != must be used for the comparison to run
B. COUNT(*) skips any row that has a NULL in any of its columns
C. NULL compared with 800 is neither true nor false, so that row is dropped
D. The cancelled row is left out automatically because its status is not paid

Q17 Spot the bug and fix it

Anand wants one row per customer with the count of orders placed in 2026, including customers with none. The tables and the query are below. The query returns only C1 2 and C3 1. Which one-line change returns C1 2, C2 0, C3 1, C4 0?

customers

customer_id
C1
C2
C3
C4

orders

order_id	customer_id	order_date
1	C1	2026-01-15
2	C2	2025-11-03
3	C1	2026-03-02
4	C3	2026-02-20

```
SELECT c.customer_id, COUNT(o.order_id) AS orders_2026
FROM customers c
LEFT JOIN orders o ON o.customer_id = c.customer_id
WHERE o.order_date >= '2026-01-01'
GROUP BY c.customer_id;
```

A. Move the date condition from WHERE into the ON clause
B. Change COUNT(o.order_id) to COUNT(*) so that the empty rows are counted
C. Change LEFT JOIN to RIGHT JOIN so that the customers side is kept
D. Add HAVING COUNT(o.order_id) >= 0 after GROUP BY

Q18 Predict the result

Against the orders table, what happens when this runs on Postgres?

```
SELECT customer_id, SUM(amount) AS total
FROM orders
WHERE SUM(amount) > 1000
GROUP BY customer_id;
```

A. It returns the customers whose total is above 1000, in the order they appear
B. It fails: an aggregate is not allowed in WHERE; the condition belongs in HAVING
C. It returns every customer, since WHERE runs before SUM exists and is quietly skipped
D. It returns no rows, since SUM is NULL at the moment WHERE is evaluated

Q19 Predict the result

Farhan's team logs every LLM call made on a support ticket. The llm_calls table below is used for Q19 and Q20. What does this query return?

call_id	ticket_id	model	prompt_tokens	completion_tokens	cost
1	T1	small	400	100	0.02
2	T1	large	1200	300	0.30
3	T2	small	800	50	0.03
4	T3	large	600	200	0.30
5	T2	small	200	20	0.01

```
SELECT ticket_id, call_id
FROM (
  SELECT ticket_id, call_id,
         ROW_NUMBER() OVER (PARTITION BY ticket_id ORDER BY call_id DESC) AS rn
  FROM llm_calls
) t
WHERE rn = 1
ORDER BY ticket_id;
```

A. (T1, 1), (T2, 3), (T3, 4)
B. Five rows, because ROW_NUMBER never removes rows
C. An error, because a window function cannot be used to filter rows
D. (T1, 2), (T2, 5), (T3, 4)

Q20 Fix the query

tickets holds one row per ticket with its category: T1 refund, T2 delivery, T3 refund. ticket_tags holds (T1, urgent), (T1, vip), (T2, urgent), (T3, vip). Anand asks for LLM spend by ticket category.

The query returns refund 0.94 and delivery 0.04, which adds to 0.98, while the cost column of llm_calls sums to 0.66. Which change gives the right totals?

```
SELECT t.category, SUM(c.cost) AS spend
FROM llm_calls c
JOIN tickets t ON t.ticket_id = c.ticket_id
JOIN ticket_tags g ON g.ticket_id = t.ticket_id
GROUP BY t.category;
```

A. Change SUM(c.cost) to SUM(DISTINCT c.cost), so each cost value is added only once
B. Add DISTINCT after SELECT, so repeated rows are removed before summing
C. Drop the ticket_tags join; the question never uses tags and it repeats T1's calls
D. Change JOIN tickets to LEFT JOIN tickets, so no ticket is lost from the totals

Section C. Numbers and reasoning

8 questions, Q21 to Q28, about 14 minutes. Rough arithmetic is enough; every answer is far from its neighbours.

Q21 Reason with numbers

Kalpa Retail revenue fell 12 percent from Q1 to Q2 while the order count rose 5 percent. Which statement about the average order value must be true?

A. It fell by about 7 percent ✓
B. It fell by about 12 percent
C. It cannot be determined without the number of customers
D. It fell by about 16 percent

Q22 Reason with numbers

Five Plus customers spent Rs 300, 350, 400, 450 and 9,000 this quarter. Meera wants one number on her slide for 'what a typical Plus customer spends'. Which number, and why?

A. 400, the median, since one bulk buyer drags the mean above the other four
B. 2,100, the mean, since it is the only number that uses every value
C. 300 to 9,000, the range, since any single number would mislead the board
D. 1,250, the midpoint between the mean and the median, as a compromise

Q23 Reason with numbers

A monsoon discount email went to 20,000 customers and produced 400 orders. A plain email went to 5,000 customers and produced 150 orders. Which group responded better?

A. The discount group, since 400 orders is more than double 150
B. The plain-email group, since 3 percent beats 2 percent ✓
C. They are level once the group sizes are allowed for
D. It cannot be judged without the revenue per order

Q24 Two statements

Statement I: A number that rises 10 percent and then falls 10 percent is back where it started.

Statement II: A number that falls 50 percent needs a rise of 100 percent to recover. ✓

A. Only I is true
B. Both are true
C. Only II is true ✓
D. Neither is true

Q25 Reason with numbers

A clinic with 50 bookings a week saw its no-show rate go from 10 percent to 14 percent this week. What is the fairest reading?

A. Two extra no-shows out of 50 sits inside normal week-to-week noise; watch a few more weeks
B. A rise from 10 to 14 percent is a 40 percent jump in the rate, which needs action this week
C. The rate has been miscalculated, since 7 out of 50 is 12 percent and not 14
D. Nothing can be said either way until a proper significance test has been run

Q26 Reason with numbers

A fraud model flags 5 percent of all transactions. It catches 90 percent of the transactions that are actually fraudulent, and fraud is 1 percent of all transactions. Of the flagged transactions, roughly what share are actually fraud?

A. About 90 percent
B. About 5 percent
C. About 1 percent
D. About 18 percent

Q27 Reason with numbers

Overall conversion fell from 5 percent to 4 percent, yet conversion rose in every single city. How can both be true?

A. They cannot both be true, since the total is just the cities added up; one figure must be wrong
B. The mix shifted toward cities with lower conversion, so the total fell as each city rose
C. Rounding in the city figures hides small falls that only show up in the total
D. Seasonality moves the total without moving any single city's own figure

Q28 Estimate

Farhan plans one LLM call per support ticket. Suppose the model bills Rs 40 per million input tokens and Rs 160 per million output tokens, and a call uses about 1,200 input tokens and 200 output tokens. At 1,000 tickets a day for 30 days, the monthly bill is closest to

A. Rs 1,440
B. Rs 6,720
C. Rs 2,400
D. Rs 24,000

Section D. Case and scenarios

6 questions. Q29 to Q34, about 10 minutes. Q29 to Q32 use one case. Read the table once, then answer all four before moving on.

The case

Meera Raghavan, CEO of Kalpa Retail, has one question for the analytics team: revenue fell from Q1 to Q2, where did it go? Kavya Nair pulled the table below. Use it for Q29 to Q32. Revenue is in Rs lakh.

Tier	Q1 customers	Q1 orders	Q1 revenue	Q2 customers	Q2 orders	Q2 revenue
Plus	10,000	30,000	360	10,000	27,000	324
Basic	40,000	60,000	420	44,000	64,000	400
Student	5,000	6,000	24	8,000	9,000	36
Total	55,000	96,000	804	62,000	100,000	760

Q29 Case: Kalpa Retail. Q1 against Q2

Which line best explains most of the fall?

A. Plus customers ordered less often at the same order value; that tier carries most of the fall
B. Basic customers paid less per order than in Q1; that tier carries most of the fall
C. Student growth pulled spending away from the Plus tier, which is why Plus orders fell by a tenth
D. Fewer customers overall placed orders in Q2, which lowered revenue across every tier

Q30 Case: Kalpa Retail, Q1 against Q2

Kavya's first hypothesis is that Plus customers are downgrading to Basic. What does the table say about it?

A. It supports it, since Basic gained 4,000 customers
B. It supports it, since the Basic order value fell
C. It weakens it, since the Plus customer count did not fall ✓
D. It cannot be judged without customer-level churn data from both quarters

Q31 Case: Kalpa Retail, Q1 against Q2

A monsoon discount was given to some Basic customers in Q2. Meera asks whether it caused the rise in Basic orders. Which comparison is the fairest one available?

A. Basic Q2 orders against Basic Q1 orders, since the before-and-after is the change in question ✓
B. Discounted Basic customers against non-discounted Basic customers, same weeks
C. Basic Q2 orders against Plus Q2 orders, since Plus had no discount
D. Basic Q2 revenue against the plan line, since the plan assumed no discount

Q32 Case: Kalpa Retail, Q1 against Q2

Before you put the Plus finding in front of Meera, which check protects the finding most?

A. Ask for the Plus tier's marketing budget for Q2 and compare it with Q1's
B. Re-add the six revenue figures in a spreadsheet to confirm that the totals match the table exactly
C. Compute the Plus tier's share of total revenue in each quarter and compare
D. Check that Q1 and Q2 have the same trading days and that no festival sat in one quarter only

Q33 Scenario: LLM intuition

Farhan's team uses an LLM to label each support ticket as refund, delivery or other. The prompt says only 'Label this ticket'. Sending the same ticket twice gives refund once and delivery the second time. Which explanation and first fix fit best?

A. Sampling above zero temperature varies the reply, and the prompt never names the labels; lower it and list the three labels with an example each
B. The model remembers the first call and changed its mind on the second; clear its conversation history before every call so each starts fresh
C. The ticket text must contain a typo or odd formatting that confuses the model; clean the data before calling again and re-check
D. The model is too small for a three-way classification; switch to the largest model available and re-run the whole batch

Q34 Scenario: LLM intuition

Kavya's prompt asks a model to rate the sentiment of each customer review from 1 to 5. On 200 reviews the ratings look plausible. Which check tells her fastest whether the ratings can be trusted?

A. Ask the model to add a confidence score to each rating and keep the high ones
B. Hand-rate a random 30 reviews herself and compare with the model's ratings
C. Run the prompt through a second model and average the two ratings for each review
D. Add 'be accurate and careful' to the prompt and re-run the whole batch

Section E. Judgment calls

6 questions. Q35 to Q40, about 5 minutes. For each situation, mark one best action and one worst action on the answer sheet. There is no neutral choice: every option has consequences.

Q35 Best and worst

You sent Anand Iyer, the Finance Controller, the quarterly revenue figure yesterday. This morning you find your query counted returned orders as sales, so the figure is about 4 percent too high, and it is already in a board deck. Mark the best action and the worst action.

A. Fix the query so next month's figure is right and say nothing about this one, since 4 percent sits inside normal variance

B. Ask a peer to re-run your query and send a correction only once they confirm the error is real

C. Send Anand the corrected figure now, with the cause and size of the error

D. Tell your reporting manager and let them decide whether Finance needs to hear about it

Q36 Best and worst

During an ungraded exercise you got the answer from an AI assistant and submitted it without understanding it. The trainer asks you to explain your approach to the room. Mark the best action and the worst action.

A. Ask the trainer to come back to you once you have had time to study the answer properly

B. Say you used an AI assistant, then walk through what you do and do not yet understand

C. Ask the person next to you, who solved it without help, to explain the approach instead

D. Present the answer as your own reasoning and improvise wherever the logic is unclear to you

Q37 Best and worst

During your group's presentation a group-mate says, in front of the panel, that your chart is misleading. Mark the best action and the worst action.

A. Say nothing now, since the panel is watching, and fix the chart after the presentation

B. Point out that their own section had errors too, so the panel hears both sides

C. Defend the chart, since you checked the numbers twice before the presentation

D. Ask what specifically misleads and offer to fix it before the presentation moves on

Q38 Best and worst

A take-home is due tonight. Your notebook runs, but the answer to the last question is wrong and you know why; the fix needs two more hours. Mark the best action and the worst action.

A. Submit on time with a note saying which part is wrong, why, and when the fix follows

B. Message the TA that you will submit tomorrow with the fix included, since correctness matters more

C. Submit as it is and say nothing, since the notebook runs and the deadline is what counts

D. Submit only the parts that are right and quietly drop the last question from the notebook